

1VPNs – Site-to-Site and Remote Access VPN

Topic: Introduction to VPNs

Main topic: Site-to-Site VPN and Remote Access VPN

1. Create Two Networks

Create two separate branch networks in Cisco Packet Tracer using two routers. Each branch has a PC and a router.

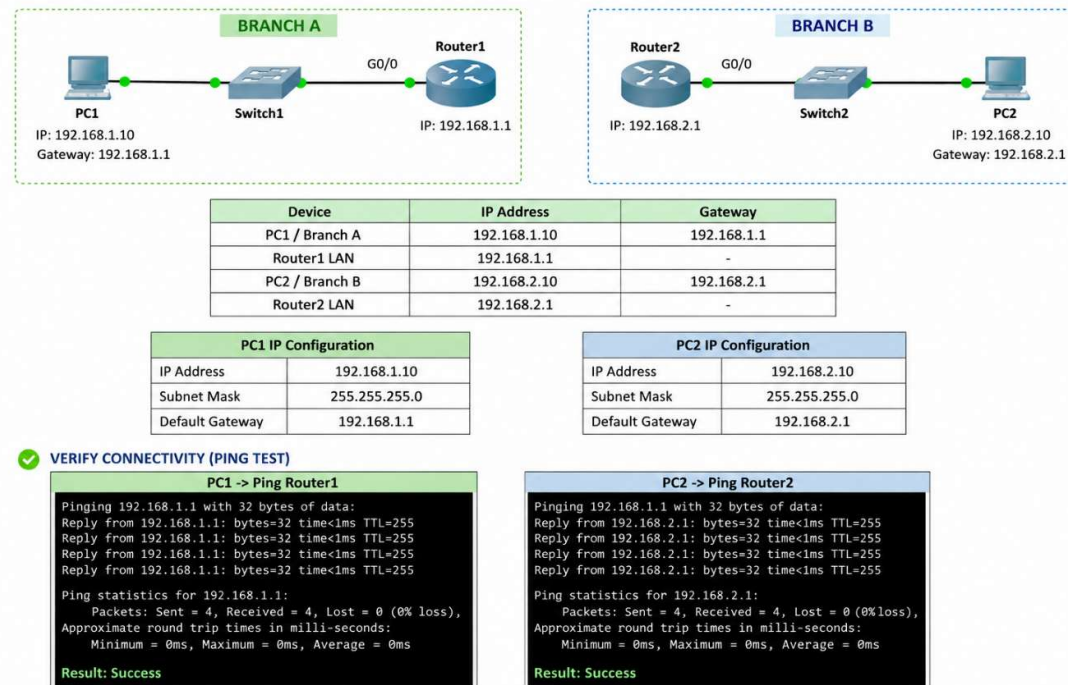
Device	IP Address	Gateway
PC1 / Branch A	192.168.1.10	192.168.1.1
Router1 LAN	192.168.1.1	-
PC2 / Branch B	192.168.2.10	192.168.2.1
Router2 LAN	192.168.2.1	-

First, verify that each PC can ping its own router.

Figure 1: Basic site-to-site VPN topology

1. CREATE TWO NETWORKS

Create two separate branch networks in Cisco Packet Tracer using two routers. Each branch has a PC and a router.



2. Configure a Basic Site-to-Site VPN

Example WAN addresses: Router1 = 10.0.0.1 and Router2 = 10.0.0.2.

Router1 commands:

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
interface tunnel 0
ip address 172.16.0.1 255.255.255.252
tunnel source g0/1
tunnel destination 10.0.0.2
exit
ip route 192.168.2.0 255.255.255.0 172.16.0.2
end
```

Router2 commands:

```
enable
configure terminal
interface g0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
interface tunnel 0
ip address 172.16.0.2 255.255.255.252
```

```
tunnel source g0/1
tunnel destination 10.0.0.1
exit
ip route 192.168.1.0 255.255.255.0 172.16.0.1
end
```

Note: If your Packet Tracer router uses different interface names, use the actual interface shown in your topology (for example, Fa0/0 instead of G0/0).

3. Remote Access VPN

A remote-access VPN allows a single user or PC to securely connect to a branch network. In Packet Tracer, configure the VPN Client on the PC and enter the VPN server/router IP, username, and password provided by the VPN configuration.

4. Site-to-Site VPN vs Remote Access VPN

Point	Site-to-Site VPN	Remote Access VPN
Use	Connects two complete networks.	Connects one remote user/device.
Advantage	Good for branch offices.	Useful for employees working remotely.
Disadvantage	Needs router/firewall configuration.	Depends on individual user/device setup.

5. Basic Configuration Steps

Create two branch LANs and assign IP addresses.

Configure the WAN/public-facing interfaces on both routers.

Create Tunnel 0 on both routers.

Set the tunnel source and destination.

Assign tunnel IP addresses.

Add routes to the opposite LAN through the tunnel.

Test the connection by pinging the PC in the other branch.

Conclusion: A site-to-site VPN connects two branch networks, while a remote-access VPN connects an individual user to a private network. GRE tunnels in Packet Tracer can be used to demonstrate the basic site-to-site connection.